Technical Architecture and Data Governance for University Hazardous Waste Management Systems: A Comprehensive Analysis of ADR 2025 Compliance, API Integration, and Regulatory Logic

1. Executive Summary and Strategic Context

The design and implementation of a digital interface for waste management within a German university context represents a complex engineering challenge that sits at the convergence of strict federal regulation, intricate chemical safety protocols, and advanced data logistics. The mandate to develop a chatbot capable of guiding university staff and students through the disposal of hazardous materials—ranging from laboratory chemicals and biological agents to lithium-ion batteries—requires a backend architecture rooted in absolute regulatory precision. Unlike consumer-facing applications where approximation is acceptable, the domain of hazardous waste (Sonderabfall) allows for no margin of error; a misclassification of a transport category or a misunderstanding of a packaging instruction can lead to severe legal penalties under the *Gefahrgutverordnung Straße, Eisenbahn und Binnenschifffahrt (GGVSEB)*, significant fines, and potential threats to human safety and environmental integrity.

This report serves as an exhaustive technical and regulatory blueprint for constructing the data layer of such a system. It synthesizes findings from the latest regulatory frameworks, specifically the *Accord relatif au transport international des marchandises Dangereuses par Route* (ADR) 2025, which entered into force on January 1, 2025.[1] The analysis identifies and evaluates the primary sources of "Source of Truth" data, specifically the newly accessible Open Data initiatives from the *Bundesanstalt für Materialforschung und -prüfung* (BAM) and the *United Nations Economic Commission for Europe* (UNECE).

A critical finding of this research is the paradigm shift in data accessibility that occurred in mid-2025. The decision by BAM to transition its "Dangerous Goods Dataservice" (Gefahrgutdatenservice) from a paid subscription model to a free, open-access service as of July 23, 2025, fundamentally alters the economic and technical viability of university-developed compliance tools.[2] This development provides a direct, authoritative pipeline to the structured data required for UN number validation, classification codes, and transport categories, eliminating the need for fragile scraping mechanisms or reliance on

third-party, potentially outdated commercial APIs.

Furthermore, this report delves into the intricate logic of the "Kleinmengenregelung" (Small Quantity Regulation) under ADR 1.1.3.6—the so-called "1000-Points Rule." It provides a granular deconstruction of the mathematical models required to implement this rule digitally, supported by specific multiplier factors and category definitions derived from the regulatory text. By integrating these calculation methodologies with the chemical inventory data from the European Chemicals Agency (ECHA) and university-specific waste guidelines (Abfall-ABCs), the proposed system can transcend simple information retrieval to become an active compliance engine.

The narrative that follows details the integration of these disparate data sources—AVV waste codes, UN transport numbers, and GHS hazard classifications—into a coherent, queryable knowledge graph. It addresses the specific "edge cases" prevalent in academic environments, such as the disposal of unknown substances, the handling of infectious clinical waste (UN 3291 vs. UN 2814), and the increasingly critical management of lithium batteries (UN 3480/3481).[3] The report concludes with a validated technical architecture recommendation that prioritizes data sovereignty, regulatory adherence, and user safety.

2. The Regulatory Framework: ADR 2025 and National Implementation

To understand the data requirements for a waste management chatbot, one must first master the regulatory environment that dictates the data's structure. In Germany, the transport of dangerous goods is governed by the GGVSEB, which implements the international ADR agreement. As of 2025, the regulatory landscape has undergone significant updates that directly impact how data must be structured and retrieved.

2.1 The Transition to ADR 2025

The ADR 2025 agreement, adopted under the auspices of the UNECE, represents the current global standard for road transport of dangerous goods.[5] For developers and data architects, the "Table A" of ADR Chapter 3.2 is the central nervous system of compliance. This table lists every dangerous good assigned a UN number, from UN 0004 (Ammonium Picrate) to the newest entries, and assigns them specific attributes that dictate their handling.[6]

The 2025 update brings critical changes. Notably, there are new special provisions and classification criteria for sodium-ion batteries, which are increasingly common in university research contexts.[7] The transitional measures for the previous regulations (ADR 2023) generally expire on June 30, 2025.[8] Therefore, any system built today must strictly utilize the 2025 datasets to ensure it is not obsolete upon deployment. The UNECE has made the full text of ADR 2025 available in English, French, and Russian [1], but the raw PDF format is insufficient

for dynamic software queries.

2.2 The Role of BAM and the "Gefahrgutdatenservice"

For a German university, the *Bundesanstalt für Materialforschung und -prüfung* (BAM) is the ultimate national authority. Their "Datenbank GEFAHRGUT" (Dangerous Goods Database) has long been the gold standard for compliance data.[9]

Historically, access to the BAM's structured data required paid subscriptions, creating a barrier for academic and open-source projects. However, a pivotal policy change has occurred: **Since July 23, 2025, the Dangerous Goods Dataservice has been available free of charge**.[2] This is a critical enabler for the proposed chatbot. The service offers data in standard formats like Excel and CSV [2], which allows for the direct ingestion of:

- **UN Numbers (UN-Nummern):** The primary keys for transport logic.
- **Proper Shipping Names (Offizielle Benennung):** Crucial for generating compliant transport documents.
- **Class and Classification Codes:** Required for determining compatibility (e.g., preventing the mixing of acids and bases).
- **Transport Categories (Beförderungskategorien):** The input variable for the 1000-Points calculation.[10]

By utilizing this official feed, the university ensures that its chatbot is disseminating data that is legally defensible. Reliance on third-party aggregators or scraping Wikipedia is insufficient for liability reasons; if a student relies on the chatbot and transports a chemical illegally, the university must demonstrate due diligence. Using the BAM feed constitutes such due diligence.

2.3 Structural Analysis of ADR Table A

To build the chatbot's database schema, we must dissect the columns of ADR Table A. The chatbot acts as a query interface for this table.

ADR Column	Data Attribute	Technical Relevance for Chatbot
Column 1	**UN Number**	**Primary Key.** All logic flows from this 4-digit identifier (e.g., UN 1993).
Column 2	**Name and Description**	**Search Vector.** Users will search for "Acetone" or "Waste Paint." The chatbot

		must map these natural language terms to the "Proper Shipping Name" found here.
Column 3a	**Class**	**UI/UX.** Determines the hazard icon (Flame, Corrosive Hand) displayed to the user.
Column 3b	**Classification Code**	**Compatibility Logic.** Used to determine if two waste containers can be packed together (e.g., F1 for flammable liquids).
Column 4	**Packing Group**	**Risk Assessment.** (I, II, III). Determines the quality of the waste canister required. High danger (I) requires more robust packaging than Low danger (III).
Column 5	**Labels**	**Visual Compliance.** Tells the user which diamond-shaped stickers must be applied to the waste container.
Column 6	**Special Provisions**	**Exception Handling.** Numeric codes (e.g., 274, 601) that trigger specific business rules in the chatbot (e.g., "If 274 is present, you must add the technical name in brackets").
Column 7a	**Limited Quantities (LQ)**	**Small Quantity Logic.**

		Defines the maximum size of inner packaging (e.g., "1 L") for simplified transport.
Column 7b	**Excepted Quantities (EQ)**	**Micro-Quantity Logic.** Codes (E0-E5) for extremely small shipments (e.g., research samples).
Column 8	**Packing Instructions**	**Operational Instructions.** Codes like P001 or IBC03. The chatbot must resolve these codes into full text instructions for the user.
Column 15	**Transport Category**	**Calculation Variable.** A single digit (0-4) used exclusively for the 1000-Points Rule calculation.
Column 15	**Tunnel Code**	**Routing.** (e.g., (D/E)). Less relevant for campus-internal transport but necessary for off-site disposal.

This structural understanding allows us to design a relational database where UN_Number is the foreign key linking hazard data, packaging requirements, and calculation variables.

3. Data Source Discovery: APIs and Datasets

The core of the user's request is "discovery"—finding the actual bits and bytes that power the system. The research identifies a tiered ecosystem of data sources, ranging from federal open data to commercial logistics APIs.

3.1 Primary Tier: Authoritative Open Data (Government & UN)

The most reliable data comes from the regulators themselves.

3.1.1 BAM Dangerous Goods Dataservice (Deutschland)

- **Access:** Via the TES portal (tes.bam.de) or dgg.bam.de.[9]
- **Format:** CSV, Excel.[2]
- **Content:** Contains the "BAM-List" which is specific to Germany. It includes approvals for tank containers and packagings that may not be in the generic UN lists.
- **Utility:** This is the backend database seed. It should be downloaded and ingested periodically (e.g., quarterly) to update the chatbot's local tables.
- **Licensing:** Free usage for non-commercial and commercial purposes under the new 2025 terms, provided proper attribution is given.[2]

3.1.2 UNECE Regulatory Files (International)

- **Access:** unece.org/transport/dangerous-goods.[1]
- **Format:** PDF (primary), Word, Excel (secondary).
- **Content:** The source text of ADR 2025 Volume 1 and Volume 2.
- **Utility:** The PDFs serve as the source for "long text" content—such as the full text of packaging instructions (e.g., the paragraph text for P001). While Table A provides the *code* P001, the UNECE PDF provides the *instructions*.
- **Processing:** Advanced RAG (Retrieval-Augmented Generation) techniques can be used to ingest these PDFs, allowing the chatbot to answer qualitative questions like "What are the construction requirements for a 4G box?" by citing the specific section of Part 6 of the ADR.[1]

3.2 Secondary Tier: Chemical Classification APIs

The chatbot needs to bridge the gap between "Substance Name" and "UN Number." This requires a chemical inventory.

3.2.1 ECHA C&L Inventory API
The European Chemicals Agency (ECHA) provides an API for the Classification and Labelling Inventory.13

- **Endpoint:** Available via the EU Open Data Portal or ECHA's dedicated API store.
- **Function:** Users input a CAS Number (e.g., 67-64-1 for Acetone) or EC Number. The API returns the GHS classification (e.g., Flam. Liq. 2, Eye Irrit. 2).
- **Integration:** While ECHA does not always provide the UN number directly, the GHS classification is the first step in determining the Transport Class. For example, if ECHA says "Flammable Liquid, Category 2," the chatbot logic can infer "Class 3, Packing Group II" is likely.[14]

3.2.2 Wikidata and Semantic Web
Wikidata is an surprisingly robust, machine-readable source for this specific domain.

- **Properties:** P2326 (UN Number), P274 (Chemical Formula), P200 (CAS Number).
- **Querying:** A SPARQL query can act as a massive translation layer.
  - *Query:* "Select all items with a UN Number and English/German labels."
  - *Result:* A mapping table of thousands of synonyms to UN numbers.[16]

- **Validation:** While community-edited, it links to PubChem and other authoritative sources, making it excellent for synonym resolution (e.g., handling "Rubbing Alcohol" vs "Isopropanol").

3.3 Tertiary Tier: Commercial and Logistics APIs

For robust, production-grade validation, commercial APIs are valuable.

3.3.1 UPS Dangerous Goods API
UPS offers a developer-friendly API that is specifically designed to validate shipments.17

- **Capability:** The "Chemical Reference Data" endpoint allows lookup by UN Number or Proper Shipping Name.
- **Output:** It returns structured JSON including the Class, Transport Category, and specific packaging limits for passenger vs. cargo aircraft.[18]
- **Relevance:** Even though the university is transporting by road (ADR), the UPS API's data on "Proper Shipping Names" is generally harmonized with UN regulations. It serves as a high-quality "spell checker" for the chatbot.[18]

3.3.2 ChemRadar
ChemRadar aggregates regulatory data from multiple jurisdictions.19

- **Feature:** Their "UN Number Search" tool provides a cross-reference between the UN number and the specific "Special Provisions" applicable in different regions (EU vs China vs US).
- **Utility:** Useful for research labs that might be importing reagents and need to understand the incoming labeling.[20]

4. The "Kleinmengenregelung" (1000-Points Rule): Deep Dive

One of the most requested features for a waste chatbot is the "1000-Points Calculation." This regulation (ADR 1.1.3.6) allows universities to transport small quantities of hazardous waste without the full burden of ADR compliance (e.g., without orange warning plates, without a fully certified ADR driver, just a 2kg fire extinguisher and general awareness training).[10]

4.1 The Mathematical Model

The calculation is a weighted sum. The "points" are calculated based on the **Transport Category** (Beförderungskategorie) of the goods.

$$\text{Total Points} = \sum_{i} (\text{Quantity}_i \times \text{Multiplier}_i)$$
Where:

- **Quantity:**

- For Solids: Mass in kg.
- For Liquids: Volume in Liters.
- For Compressed Gases: Water capacity of the cylinder in Liters.

The **Multiplier** depends strictly on the **Transport Category** (Column 15 of Table A).

4.2 Transport Categories and Logic Table

The chatbot must contain this hardcoded logic table, derived from ADR 1.1.3.6 [21]:

Transport Category (BK)	Multiplier (Faktor)	Significance & Examples	Maximum Quantity for Exemption (1000 Pts)
0	**∞ (Prohibited)**	Extremely dangerous. Transport in "Kleinmengen" is not allowed. Examples: Infectious substances Cat A (UN 2814), Toxic Gases (UN 1005).	**0**
1	**50**	Very High Danger. Examples: Explosives (1.1), Toxic substances PG I (UN 2810), Arsenic.	**20** (kg/L)
2	**3**	High/Medium Danger. Examples: Flammable Liquids PG II (Acetone, Ethanol), Lithium Batteries (UN 3480).	**333** (kg/L)

3	**1**	Low Danger. Examples: Flammable Liquids PG III (Diesel, Heating Oil), Acids/Bases PG III.	**1000** (kg/L)
4	**0**	Very Low Danger. Examples: Empty uncleaned packaging (UN 3509), limited quantities.	**Unlimited**

4.3 Implementation Scenarios and Edge Cases

The chatbot must handle complex scenarios typical of a university waste run.

Scenario A: Mixed Load

- **Load:** 50 Liters of Acetone (UN 1090) and 10 kg of Lithium Batteries (UN 3480).
- **Lookup:**
  - Acetone: Class 3, PG II $\rightarrow$ Transport Category **2**.
  - Batteries: Class 9 $\rightarrow$ Transport Category **2**.
- **Calculation:**
  - Acetone: $50 \times 3 = 150$ points.
  - Batteries: $10 \times 3 = 30$ points.
  - Total: 180 points.
- **Decision:** $180 \le 1000$. **Result:** "Exemption Applies. No orange plates needed."

Scenario B: The "Toxic" Trap

- **Load:** 5 Liters of a highly toxic synthesis by-product (UN 2810, Toxic Liquid, Organic, N.O.S., Packing Group I).
- **Lookup:** Packing Group I implies Transport Category **1**.
- **Calculation:** $5 \times 50 = 250$ points.
- **Decision:** $250 \le 1000$. **Result:** Exemption Applies.
- **Analysis:** Even though the quantity is small (5L), it consumes a quarter of the "points" budget. If the researcher added 250L of Acetone ($250 \times 3 = 750$), the total would be 1000 ($250 + 750$), which is the exact limit. Any more would trigger full regulation.

Scenario C: Empty Packaging (Ungereinigte Leerverpackungen)

- **Load:** 50 empty plastic canisters that contained Acetone.

- **Logic:** Empty packaging often falls under Transport Category 4 (Multiplier 0).
- **Result:** 0 Points. They do not count towards the limit.[10]
- **Chatbot Output:** "These empty containers do not add to your ADR points score, provided they are not leaking."

4.4 Exemptions vs. Limited Quantities (LQ)

The chatbot must distinguish between the **1000-Points Rule** (1.1.3.6) and **Limited Quantities** (3.4).

- **Limited Quantities (LQ):** Refers to the packaging *size* (e.g., small bottles in a box). If goods are packed in "Limited Quantities" (e.g., 1L bottles of Ethanol in a box marked with the LQ diamond), they are **exempt from the points calculation entirely**.[10]
- **Chatbot Logic:**
  - *User Question:* "I have 50 boxes of Ethanol. Each box has 10 x 1L bottles. They have the LQ diamond."
  - *Chatbot Answer:* "Since these are packed as Limited Quantities (LQ), they do not count towards the 1000 points. You can transport them without an ADR driver license, provided the gross weight exceeds 8 tonnes marking requirements apply."

This nuanced distinction is vital. A naive calculator would sum up the 500L of Ethanol ($500 \times 3 = 1500$) and falsely warn the user that they are non-compliant. The chatbot must ask: "Is the packaging marked with the LQ diamond (black corners)?"

5. Bridging the AVV to UN Gap: Waste Classification

In the university context, the primary identifier for waste is the **AVV Code** (Abfallverzeichnis-Verordnung), a 6-digit code based on the European Waste List. However, transport requires a **UN Number**. There is no official 1:1 mapping between these two systems because one is based on *source* and the other on *hazard*.[23]

5.1 The Mapping Disconnect

- **AVV Logic:** "Where did this come from?" (e.g., 14 06 03* - Other solvents and solvent mixtures).
- **ADR Logic:** "What is the danger?" (e.g., UN 1993 - Flammable Liquid, or UN 2924 - Flammable Liquid, Corrosive).

A single AVV code like 16 05 06* (Lab chemicals consisting of hazardous substances) can map to hundreds of different UN numbers depending on whether the chemical is acidic, toxic, flammable, or oxidizing.

5.2 Heuristic Mapping Strategy

To solve this, the chatbot must employ a **Heuristic Decision Tree** utilizing university-specific

"Abfall-ABCs" (Waste Guides) as training data.

Case Study: Clinical Waste

- **User Input:** "I have a bag of petri dishes with bacteria cultures."
- **AVV Lookup:** Likely 18 01 03* (Wastes whose collection and disposal is subject to special requirements in order to prevent infection).
- **ADR Decision:**
  - *Question:* "Is this waste Category A (high risk, e.g., Ebola, Anthrax) or Category B (routine cultures)?"
  - *Path A:* Category A $\rightarrow$ **UN 2814** (Infectious substance, affecting humans). Transport Category 0 (Prohibited in Kleinmengen).
  - *Path B:* Category B / Routine $\rightarrow$ **UN 3291** (Clinical Waste, Unspecified, N.O.S.). Transport Category 2.
- **Source:** This logic is explicitly detailed in university guides like the *Abfallhandbuch Uni Hamburg* [24] and *RWTH Aachen* protocols.[25]

Case Study: Laboratory Packs (Lab Chemicals)

- **User Input:** "Old bottles of various acids."
- **AVV:** 16 05 06*.
- **ADR Decision:**
  - Mixed acids often go into a "collection drum" (Sammelbehälter).
  - Common UN Number for mixed acids: **UN 3264** (Corrosive liquid, acidic, inorganic, n.o.s.) or **UN 1789** (Hydrochloric acid) if pure.
  - The chatbot should recommend: "If mixed inorganic acids, use UN 3264. If mixed organic acids, use UN 3265."

5.3 Data Ingestion from "Abfall-ABCs"

Universities publish detailed PDF guides (Abfall-ABC) that serve as "Rosetta Stones" for this mapping.

- **Strategy:** The research agent should scrape/ingest these PDFs from major German universities (FU Berlin, TU Munich, Uni Duisburg-Essen).[24]
- **Data Extraction:**
  - Extract tables mapping "Waste Description" $\rightarrow$ "AVV" $\rightarrow$ "UN Number".
  - Example from TU Munich: "Fixierbäder" (Fixing baths) $\rightarrow$ AVV 09 01 04 $\rightarrow$ UN Number (often not dangerous goods if non-hazardous, or specific UN if silver content is high).[26]

6. Packaging Requirements and Logistics

Once the UN number and quantity are determined, the user needs to know *how* to pack the

waste. ADR uses "Packing Instructions" (Verpackungsanweisungen) represented by codes like P001, P002, IBC02.

6.1 Decoding P-Codes

The chatbot must translate the alphanumeric code into physical reality.

- **P001 (Liquids):** "Use a combination packaging. Inner: Glass bottle max 10L. Outer: Fibreboard box (4G) max 400kg."
- **P002 (Solids):** "Use a drum (1A2) or box (4G)."

6.2 University-Specific Container Inventory

Universities standardise their waste containers to simplify logistics. The chatbot should map regulatory P-codes to the specific "Article Numbers" in the university's procurement system.

- **Liquid Chemicals:** usually go into **10L PE-Canisters** (Blue or White). These correspond to UN-approved single packagings (e.g., 3H1 plastic jerricans).
- **Solid Waste:** usually goes into **ASP-Behälter** (large plastic or steel bins with clamp lids).
- **Battery Collection:** Specific drums with vermiculite (fire suppressant).

Chatbot Output Example:

> "For UN 1993 (Waste Solvent), standard packaging is required.
>
> - **Regulatory Code:** P001.
> - **University Standard:** Use the **Blue 10L Safety Canister** (Stock #12345).
> - **Labeling:** Apply the 'Flammable Liquid' (Class 3) diamond and the 'UN 1993' sticker."

6.3 Labelling and Documentation

The chatbot must also generate the text for the "Beförderungspapier" (Transport Document).

- **Format:** UN Number, Proper Shipping Name, Class, Packing Group, (Tunnel Code).
- **Example:** UN 1993, FLAMMABLE LIQUID, N.O.S. (Acetone, Ethanol), 3, II, (D/E).
- **Note:** The technical names in brackets are required by **Special Provision 274** (Column 6 of Table A). The chatbot must check Col 6 and prompt the user: "Please specify the two main hazardous components for the document."

7. Technical Architecture and API Integration

To synthesize all these requirements, I propose a "Hybrid Knowledge Architecture."

7.1 Database Schema (Relational + Vector)

1. **Relational Database (PostgreSQL):**
    - **Table adr_master:** Sourced from BAM CSV. Columns: UN, PSN, Class, PG,

TransportCategory, TunnelCode, PackingInst.
    - **Table ghs_chemicals:** Sourced from ECHA/Wikidata. Columns: CAS, EC, Name, UN_Mapping (Likely).
    - **Table university_inventory:** Columns: ContainerType, StockNumber, P_Code_Compatibility.
2. **Vector Database (Pinecone/Milvus):**
    - Ingest the full text of ADR 2025 (PDFs) and University Abfall-ABCs.
    - Enables RAG queries for qualitative questions ("Can I leave the lid open if it's fuming?").

7.2 API Integration Pattern

- **Primary Logic:** Local SQL queries against the adr_master table (fast, deterministic, offline-capable).
- **Validation:** Use **UPS Dangerous Goods API** (ChemicalReferenceData) as a secondary check during the "ingestion phase" to ensure the local database remains accurate.[17]
- **Search:** Use **MeiliSearch** or **Elasticsearch** indexed on the "Proper Shipping Names" (Column 2) and synonyms from Wikidata to allow fuzzy matching (e.g., user types "Alkohol" $\rightarrow$ finds "UN 1170 ETHANOL").

7.3 The "1000-Points" Microservice

A dedicated calculation module should be exposed:

- **Input:** JSON Array of waste items {un_number, quantity_l_kg}.
- **Process:**
    1. Fetch transport_category for each UN from adr_master.
    2. Map to multiplier (0, 1, 3, 50).
    3. Compute sum = quantity * multiplier.
- **Output:** {total_points: 450, is_exempt: true, mandatory_equipment: ["2kg Fire Extinguisher"]}.

8. Conclusion

The construction of a German university waste disposal chatbot is a task of significant gravity, requiring a rigorous synthesis of federal open data and complex logistical logic. The release of the **BAM Dangerous Goods Dataservice** as a free resource in 2025 provides the necessary foundation, allowing for a compliant, deterministic backend that was previously cost-prohibitive.

By layering the **1000-Points calculation logic** and a **University-specific waste mapping** heuristic on top of this official data, the system can guide users safely through the regulatory maze. This report provides the exhaustive data map—identifying the exact columns, multipliers, and API endpoints—required to build a system that protects both the university's

legal standing and the physical safety of its students and staff.

The path forward is clear: Move away from ad-hoc lists and build a robust SQL-based engine fueled by the BAM and UNECE datasets, augmented by the semantic richness of vector-embedded university guidelines. This "Hybrid Knowledge" approach ensures that the chatbot is not just a conversational interface, but a verified compliance officer in digital form.

Summary of Key Data Sources

Source	Content Relevance	Access Method	Status
BAM Gefahrgutdatenservice	**Core ADR 2025 Data (Table A)**	Download (CSV/Excel) from tes.bam.de	**Free (New in 2025)**
UNECE Dangerous Goods	**Full Regulatory Text (Packaging Instructions)**	PDF Download from unece.org	Open Access
ECHA C&L Inventory	**Chemical Name to Hazard Mapping**	API / Bulk Download	Open Access
UPS Developer API	**Validation of Shipping Names**	REST API (ChemicalReferenceData)	Free (Requires Key)
University Abfall-ABCs	**AVV to UN Heuristic Mappings**	PDF Scraping (e.g., Uni Hamburg, RWTH)	Publicly Available
Wikidata (SPARQL)	**Synonyms & Multilingual Labels**	SPARQL Query (P2326)	Open Data

This architecture ensures a "Safety First" approach, leveraging the best available data to mitigate risk in a high-stakes environment.

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